

SULTAN MAHMUD

Backend & Infrastructure Engineer

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PROFESSIONAL SUMMARY

Backend & Infrastructure Engineer specializing in **distributed systems** and **Kubernetes infrastructure**. Built production-grade portfolio projects: **ML-enhanced K8s autoscaler (60% cost reduction)**, **polyglot microservices (sub-1ms latency)**, and infrastructure serving **5,000+ users**. Deep expertise in **AWS, Python, Go, observability stacks** (OpenTelemetry, Tempo, Grafana). Seeking full-time **engineering** role.

EXPERIENCE

BACKEND ENGINEER & PRODUCT BUILDER | *Remote*

August 2024 - Present

Building production systems to demonstrate platform engineering capabilities while actively seeking full-time opportunities

- Architected and deployed 5 production-grade applications serving **5,000+ real users** across e-commerce, fintech, and SaaS domains, managing **11+ EC2 instances** with **99.9% uptime** through **multi-AZ AWS infrastructure**
 - Built **ElastiKube: ML-enhanced Kubernetes autoscaler** with **4-layer intelligent scaling** (time-aware, flash sale detection, Prophet forecasting), achieving **60% cost reduction** through **event-driven Lambda architecture** and **multi-AZ high availability**
 - Engineered **polyglot URL shortener**: **Go `redirect_service`** (sub-1ms latency), **Python `create_service`**, with **comprehensive observability stack** (OpenTelemetry, Tempo, Grafana, Loki) and **circuit breaker fault tolerance**
 - Deployed **production infrastructure** using **Pulumi IaC** and **Ansible** across **multi-AZ VPC**, implementing **automated failover, distributed tracing, and zero-downtime CI/CD pipelines**
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BACKEND DEVELOPER | *Cooking Station, Dhaka*

June 2024 - August 2024

- **Streamlined operations** for 200+ users by designing **role-based admin dashboard** with **real-time meal analytics**, reducing **manual subscription management overhead**
 - **Eliminated 40% of manual effort** in account management through **automated balance updates** and **error-prevention systems**
 - Built **production-ready meal scheduling system** using **cron jobs** with **configurable time boundaries** and **daily user listing functionality**
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NOTABLE PROJECTS

ELASTIKUBE: PRODUCTION K3S AUTOSCALER FOR AWS | [Demo](#)

Challenge: Build production-grade intelligent autoscaler with ML-based predictive scaling for K3s clusters on AWS

Impact: Event-driven Lambda architecture with 4-layer intelligent scaling achieving 60% cost reduction (\$200→\$111-\$149/month), multi-AZ high availability, and proactive pre-scaling before traffic spikes

Technical Achievements:

- Architected **4-layer intelligent autoscaling**: (1) **Data Collection**, (2) **Time-Aware Scaling** (peak/off-peak: 85%/60% vs 60%/40%), (3) **Flash Sale Detection** (>30% spike in 2min), (4) **Prophet ML forecasting CPU 15min ahead**
- **Built ML training pipeline** with **Kubernetes CronJob** for weekly automated retraining, **feature engineering** (cyclical encoding, lag features), **cross-validation**, and **backtesting**
- **Implemented distributed locking** via **DynamoDB conditional writes**, **Write-Ahead Log (WAL)** for crash recovery, and **EventBridge orchestration** for fault-tolerant scaling
- Engineered **multi-AZ high availability** with round-robin distribution across 3 AZs, **LIFO scale-down** with **kubectl drain** via **SSM**, and single **NAT Gateway** optimization
- Deployed comprehensive **observability**: 17 **CloudWatch alarms** (CRITICAL/WARNING), **Prometheus graceful degradation**, fixed **LogGroups**, **DLQ monitoring**, and **SNS alerting**
- Built **spot instance support** with automatic **On-Demand fallback** handling 3 **AWS capacity exceptions** for **70% cost savings**
- Implemented comprehensive **security hardening**: **IAM least-privilege policies**, **security group lockdowns**, automated **SSL certificate renewal**, and **secrets management** via **AWS Secrets Manager**

Key Innovation: **Hybrid reactive+proactive scaling** combining **time-aware thresholds**, **flash sale detection**, and **ML forecasting** for **SLA-driven pre-scaling**

ULTRA-FAST URL SHORTENER MICROSERVICE

Challenge: Design distributed URL shortener with independent service scaling and sub-10ms redirects

Impact: Achieved 83% latency reduction (10ms → sub-1ms) by porting redirect service to Go, deployed comprehensive observability stack with Tempo tracing and Loki log aggregation

Technical Achievements:

- **Architected** polyglot microservices: **Go redirect_service (95% read traffic, sub-1ms)**, **Python create_service** (write-heavy), **Celery worker_service** with independent **scaling**.
- **Implemented** high-performance Go redirect service using **Chi router**, internal/clean architecture, **circuit breaker pattern** (Closed → Open → Half-Open), and **cache-aside pattern** with **30-minute Redis TTL**.
- **Engineered complete observability:** **OpenTelemetry instrumentation** with **OTLP collector**, **Tempo** distributed tracing, **Loki** log aggregation via **Promtail**, and **Grafana** dashboards for end-to-end service visibility.
- **Built repository pattern** with abstract base classes for **PostgreSQL/MongoDB**, **optimized key allocation** using **SELECT FOR UPDATE SKIP LOCKED** for **race-free operations**, and parameterized **bulk inserts (100K+ keys)**.
- **Implemented intelligent key pre-population** using **Celery workers** maintaining a pool of unused **short URL keys** for **instant URL creation** without database latency.
- **Engineered production resilience:** **PgBouncer connection pooling** achieving **53% overhead reduction** with **exponential backoff retries** and database **timeout protection**.
- **Built comprehensive testing infrastructure:** **multi-database mocking** (SQLite, mongomock, fakeredis), **async pytest** framework, **httpx API client testing**, and isolated test environments.
- **Deployed on K3s/AWS** via **Pulumi IaC**, **Ansible** configuration, and **GitHub Actions CI/CD** with automated infrastructure provisioning.

Key Innovation: Polyglot architecture demonstrating pragmatic technology selection—**Go** for performance-critical paths (**sub-1ms redirects**), **Python** for business logic, enabling **independent service scaling** without rewrites.

TECHNICAL SKILLS

- **CLOUD & INFRASTRUCTURE:** AWS(EC2, S3, VPC, Lambda, EventBridge, CloudWatch, SNS, SQS), **Docker**, Kubernetes(K3s), Pulumi, Ansible
 - **SECURITY & COMPLIANCE:** IAM policies, Security Groups, SSL/TLS, Secret Management, Vulnerability Scanning
 - **DEVOPS & CI/CD:** GitHub Actions, GitLab CI/CD (familiar), Docker Compose, Nginx, Load Balancing
 - **OBSERVABILITY & MONITORING:** OpenTelemetry (Distributed Tracing), Prometheus, Grafana, Tempo
 - **LANGUAGE & FRAMEWORKS:** Python, Golang, JavaScript, **FastAPI**, Django, DjangoREST
 - **DATABASE & CACHING:** PostgreSQL, Redis, MongoDB, SQLAlchemy, PgBouncer (Connection Pooling), Alembic (Migrations)
 - **API DEVELOPMENT:** Swagger/OpenAPI, RESTful API, Authentication (JWT), Rate Limiting, CORS
 - **MESSAGE QUEUING & ASYNC:** Celery(Worker, Beat, Flower), Redis, RabbitMQ, Async/Await
 - **ARCHITECTURE & DESIGN:** Microservices, Distributed Systems, High Availability, Fault Tolerance, Load Balancing, Caching Strategies, Database Optimization
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EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE & ENGINEERING

Daffodil International University, Dhaka

September 2017 - September 2023

EXTRACURRICULAR ACTIVITIES & COMMUNITY ENGAGEMENT

- **Quora Bengali:** Maintain A Dedicated Blog On **Quora** Offering Insights And Solutions To Inquiries Spanning Technology, Culinary Arts, And Health In Bengali. Accumulated **Readership** Nearing **200,000** With Nearly **200 Followers**.
 - **Medium Technical Blog:** Published 2 in-depth technical articles (1,000+ words each) discussing distributed systems, backend architecture & engineering decisions and trade-offs
 - **YouTube Channel:** Founded And Managed The Youtube Channel "**I.T. Darshonik**" Producing Over **40 Instructional Videos** In Bengali, Aimed At Bridging The Language Gap In Tech Education.
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AWARDS & ACHIEVEMENTS

- **DIU Take-Off Programming Contest, Spring 2018:** Ranked **6th/300+ participants** (Top 2%)
- **Competitive Programming:** Solved 500+ problems across BeeCrowd, LightOJ, HackerRank, LeetCode.
- **Proficiency:** Dynamic Programming, Graph Theory, Greedy Algorithms, Binary Search